AI-Assisted Software Development

A Guide to the Intelligent Features of AI Dev Tools

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Github Copilot Overview

From Microsoft in collaboration with OpenAI
Originally a plugin for VS Code to provide TAB auto-completion
Grew from Microfoft's ownership of Github and its many codebases

1. **Core Capabilities**
    - AI pair programmer
    - Integrates directly into IDE
    - Also accessible online (github.com)
    - Integrates with github workspaces
    - Code Citations (for code it finds in Github)

3. **Key Features**
    - Real-time code suggestions
    - Comment-to-code generation
    - Context from entire file
    - Supports 12+ languages
    - Available in VS Code, JetBrains, Vim, etc.

- Allows choice of models not just those from OpenAI,
    - Copilot team prefer Claude models.
- Free tier available, limited by usage (number of chats / month..)

Cursor Overview

Cursor is an AI-first code editor designed to enhance software development productivity through a suite of intelligent features.
It uses a range of LLMs depending on purpose / complexity of the query.

1. **Core Capabilities**
   - Fork of VS Code with AI deeply integrated
   - Can understand entire codebases
   - Conversational interface

3. **Key Features**
   - Codebase-wide context
   - Multi-file edits
   - Terminal integration
   - Checkpoints (accept/revert)
   - Privacy-first approach
     including "No Train" agreement with OpenAI

- Optimises token usage for cost-saving.
- Free tier available, time-limited (number of days usage).

Windsurf Overview

A popular AI-powered IDE By Codium who developed coding LLMs.
Developed from their predecessor product, the Codeium VS Code extension.
Implemented as a Fork of VS Code, subsequently acquired by OpenAI.

1. **Core Capabilities**
    - AI coding assistant with agentic capabilities
    - Can work autonomously on tasks
    - Multi-step problem solving
    - Built-in code execution

3. **Key Features**
    - Agentic workflows
    - Can run tests and iterate
    - Understands project structure
    - Long-running tasks
    - Integrated debugging

Claude Code Overview

Originally a command-line tool Developed by Anthropic for their own use.
Later implemented integration with IDEs.
Focusses on most effective and powerful use of LLMs so can be costly in token usage.

1. **Core Capabilities**
    - Powerful terminal/shell pipeline integration
    - Simple user interface
    - Strong reasoning capabilities
    - Focus on security and accuracy

3. **Key Features**
    - File operations and code search
    - Extended context windows
    - Thoughtful code generation
    - Tool integration

- No "Checkpoint" feature so make small frequent git commits.
- Payment plans can be more cost-effective than API key for extensive usage.
- Defaults to using one of Anthropic's models but can be configured for others.

OpenAI Codex and Gemini Code Assist

1. **OpenAI Codex CLI**
   - Command-line interface to Codex
   - Natural language to code
   - Good for scripting and automation

3. **Gemini Code Assist**
   - Google's coding assistant
   - Integrates with Google Cloud
   - Strong multi-modal capabilities
   - Good for cloud-native development

Choosing the Right Tool

Consider:

1. **Your Workflow**
   - IDE vs. terminal preference
   - Solo vs. team development
   - Project size and complexity
   - Cost and availability

3. **Technical needs**
   - Language support
   - Context window size
   - Codebase integration
   - Privacy requirements

- In practice most developers use multiple tools for different tasks

Cursor Code Generation

- **Tab**: Robust autocomplete function predicting your next edits
- **Multi-Line Edits**: Simultaneous modifications across multiple lines
- **Smart Rewrites**: Automatically corrects typographical errors and syntax issues
- **Instant Apply**: Implement code suggestions from the chat directly into your codebase
- **Cursor Prediction**: Anticipates your next cursor position for seamless navigation

Chat

- Integrates an AI assistant that understands your codebase
- Ask questions like "Is there a bug here?" and receive context-aware responses
- **Codebase Answers**: Query your entire codebase for relevant snippets or information

Querying Your Codebase with Cursor

- **Use Images**: Drag and drop images into the input box for visual context
- **Reference Your Code**: Use "@" to reference specific code files or symbols
- **Ask the Web**: Retrieve up-to-date information from the internet
- **Use Documentation**: Reference popular libraries or add your own documentation

Cursor Inline Chat (Ctrl K)

- Edit and write code with AI assistance
- **Fast Edits**: Describe desired changes for quick implementation
- **Terminal Ctrl K**: Convert plain English instructions into terminal commands
- **Quick Questions**: Get immediate answers to specific code-related queries

Intelligent Code Completion

- Context-aware code suggestions to expedite coding and reduce errors
- **Access Method**: Cursor offers code completions as you type
- **Keyboard Shortcuts**:
  - Accept Suggestion: Press Tab
  - Reject Suggestion: Press Esc
  - Partial Accept: Press Ctrl (Windows) or ⌘ (Mac) + →

Inline Code Generation and Editing (Cmd K)

- Cursor can generate or modify code snippets using natural language prompts
- **Access Method**: Highlight the code segment you wish to modify
- **Keyboard Shortcuts**:
  - Open Inline Editing: Ctrl + K (Windows) / ⌘ + K (Mac)
  - Apply Changes: Ctrl + Enter (Windows) / ⌘ + Enter (Mac)
  - Cancel/Delete Changes: Ctrl + Backspace (Windows) / ⌘ + Backspace (Mac)

AI Chat Interface

- Engage in conversations with an AI for explanations, debugging, and code suggestions
- **Access Method**: Open AI Sidebar and navigate to Chat
- **Keyboard Shortcuts**:
  - Open Chat: Ctrl + L (Windows) / ⌘ + L (Mac)
  - Add Code to Chat: Ctrl + L (Windows) / ⌘ + L (Mac)

Cursor Plan mode

- Plans proposed steps without executing them
- Can produce Checklist and Markdown documentation
- Choose "Plan" in the "Agent / Ask" dropdown

Codebase Querying with @ Symbols

- Reference specific files, functions, or variables to provide context in AI interactions
- **Access Method**: Type @ followed by the name of the file, function, or variable
- **Examples**:
    - @filename to reference a file
    - @functionName to reference a function
    - @variableName to reference a variable

Instant Apply

- Apply code suggestions from the AI chat directly into your codebase
- **Access Method**: Click the Apply button above any code block in the chat

Codebase Answers

- Pose questions about your codebase, and Cursor will search through relevant files to provide answers
- **Access Method**: In the chat interface, type your question and use the @codebase tag

Cursor Bugbot

Was called "AI Reviewer" then "Bug Finder"

Compares to previous version in Git

- Receive real-time code reviews to identify potential bugs and optimize code quality
- **Access Method**: After making code changes, access the Bugbot feature through the Command Palette or designated UI button
  See icon in Left-side Activity bar
- Can link to github / gitlab to do real-time reviews of PRs
- Can be costly in token usage

Recap

- Cursor is an AI-powered code editor designed to enhance developer productivity
- Features include intelligent code assistance and seamless integration of AI capabilities
- For a comprehensive list of features and updates, visit Cursor's official Features Page